SQL Analysis Report

D2C CUSTOMER INTELLIGENCE ENGINE

Customer Segmentation • Retention Strategy • Promotional Intelligence

Team

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About This Report

The D2C Customer Intelligence Engine is a SQL-driven analytical framework designed to uncover deep behavioral patterns, loyalty signals, and promotional dependency across a direct-to-consumer customer base.
This report presents five core analyses addressing the most critical questions in D2C customer strategy, plus an extended bonus analysis on seasonal behavior.

What's Covered

1. Loyal vs Promo Buyers
   Who are genuinely loyal customers vs. those who only buy with discounts?

2. Customer Value Tier Breakdown
   What behavioral patterns predict high customer value over time?

3. Geographic Demand Intelligence
   Which geographies signal organic demand vs. discount-driven volume?

4. Promotional Sunset Plan
   How should the brand restructure its promotional strategy to protect margins?

5. Ideal Customer Profile
   What does the brand's best customer actually look like?

\+ Season x Category -Tenure Analysis (Additional Analysis -Beyond Scope)
  Which seasons and categories attract new vs. established repeat customers?

Dataset & Methodology

1. **Source Table:** “refined_customer_intelligence_engine_output” renamed to → d2c_customers

2. **Key Fields:** Master_Customer_Segment, Loyalty_Score_A/B, Promo_Dependency_Score, Purchase_Amount_USD, Customer_Value_Tier, Location, Gender, Season, Category

3. **Approach:** Aggregated SQL analytics using AVG, COUNT, SUM, CASE logic, window partitioning, and conditional segmentation thresholds.

Threshold Design Philosophy:

- Thresholds were intentionally designed as strategic heuristics rather than statistically optimized cutoffs.

  The objective was:

- behavioral interpretability
- founder-level usability
- segmentation clarity
- retention decision support

  instead of unnecessary scoring complexity that reduces interpretability.

The objective was to balance analytical usefulness with business interpretability for non-technical stakeholders.

1

Loyal vs Promo Buyers

Who are genuinely loyal customers vs. those who only buy with discounts?

```sql
SELECT
    Master_Customer_Segment,
    COUNT(*) AS Total_Customers,
    ROUND(AVG(Promo_Dependency_Score), 2) AS Avg_Promo_Dependency,
    ROUND(AVG(Loyalty_Score_A), 2) AS Avg_Behavioral_Loyalty,
    ROUND(AVG(Loyalty_Score_B), 2) AS Avg_Commercial_Loyalty,
    ROUND(AVG(Purchase_Amount_USD), 2) AS Avg_Customer_Spend
FROM d2c_customers
GROUP BY Master_Customer_Segment
ORDER BY Avg_Behavioral_Loyalty DESC;
```

Query Result:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Master_Customer_Segment	Total_Customers	Avg_Promo_Dependency	Avg_Behavioral_Loyalty	Avg_Commercial_Loyalty	Avg_Customer_Spend
Premium Loyalists	1420	0.00	0.8	0.63	60.19
Emotionally Loyal	331	0.00	0.67	0.44	58.21
Moderate Customers	472	0.00	0.6	0.4	61.29
Discount Survivors	1640	1.00	0.5	0.55	59.12

Key Insights:

- Premium Loyalists showed almost 0 promo dependency, meaning they purchase because they genuinely value the brand, not because of discounts.
- Discount Survivors showed significantly higher promo dependency, indicating stronger sensitivity toward discounts and promotional incentives.
- Behavioral loyalty and commercial loyalty scores clearly separated loyal customers from price-sensitive customers.
- Average spending patterns alone were not sufficient to distinguish genuinely loyal customers from promotion-driven buyers.
- Some promo-dependent customers may still represent commercially valuable seasonal revenue opportunities.
- The strategic objective is not to eliminate promotions entirely, but to reduce unnecessary discount dependency while protecting long-term margin quality.

Business Understanding:

The company should not waste discounts on already loyal customers. Instead:

- Reward loyal customers with exclusive perks.
- Create a gradual discount reduction strategy for discount-dependent buyers

2 Customer Value Tier Breakdown

What behavioral patterns predict high customer value over time?

```sql
SELECT
    Customer_Value_Tier,
    ROUND(AVG(Purchase_Intensity_Score), 2) AS Avg_Purchase_Intensity,
    ROUND(AVG(Review_Satisfaction_Score), 2) AS Avg_Satisfaction,
    ROUND(AVG(Promo_Dependency_Score), 2) AS Avg_Promo_Dependency,
    ROUND(AVG(Previous_Purchases), 1) AS Avg_Previous_Purchases,
    COUNT(*) AS Customer_Count
FROM d2c_customers
GROUP BY Customer_Value_Tier
ORDER BY Avg_Purchase_Intensity DESC;
```

Query Result:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Customer_Value_Tier	Avg_Purchase_Intensity	Avg_Satisfaction	Avg_Promo_Dependency	Avg_Previous_Purchases	Customer_Count
Mid Value	0.51	0.75	0.43	25.4	1288
High Value	0.51	0.75	0.42	25.3	1282
Low Value	0.51	0.75	0.43	25.3	1293

Key Insights:

- High, Mid, and Low value customers showed very similar:
  - Purchase intensity
  - Satisfaction score
  - Promo dependency
- This means customer value alone does not predict loyalty.
- Some high-spending customers may still be highly discount dependent.

Business Understanding:

Revenue alone is not enough to identify the best customers.
The company should combine:

- Customer value
- Loyalty behavior
- Promo dependency

for smarter targeting

3 Geographic Demand Intelligence

Which geographies signal organic demand vs. discount-driven volume?

```sql
SELECT
  Location,
  COUNT(*) AS Total_Customers,
  ROUND(AVG(Purchase_Amount_USD), 2) AS Avg_Spend,
  ROUND(AVG(Promo_Dependency_Score), 2) AS Avg_Promo_Dependency,
  SUM(CASE WHEN Master_Customer_Segment = 'Premium Loyalists'
      THEN 1 ELSE 0 END) AS Premium_Loyalists,
  ROUND(100.0 * SUM(CASE WHEN Master_Customer_Segment
    = 'Premium Loyalists' THEN 1 ELSE 0 END) / COUNT(*), 1) AS Premium_Pct
FROM d2c_customers
GROUP BY Location
ORDER BY Avg_Spend DESC, Avg_Promo_Dependency ASC;
```

Query Result:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Location	Total_Customers	Avg_Spend	Avg_Promo_Dependency	Premium_Loyalists	Premium_Pct
Alaska	72	67.60	0.40	30	41.7
Pennsylvania	74	66.57	0.45	30	40.5
Arizona	65	66.55	0.34	27	41.5
West Virginia	79	63.86	0.48	25	31.6
Washington	73	63.33	0.44	18	24.7

Key Insights:

- Locations like Arizona and Alaskashowed:
  - High average spending
  - Lower promo dependency
- These regions may indicate relatively stronger organic brand demand compared to heavily discount-driven regions.
- Some regions showed decent revenue but high discount reliance, meaning sales are fragile.
- High-spend regions with lower promotional dependency may represent stronger long-term expansion opportunities compared to regions driven primarily by discount-sensitive purchasing behavior.

Business Understanding:

The company should:

- Invest more marketing budget in organically strong regions.
- Reduce discount dependency carefully in weaker regions.
- Use geography-based targeting strategies.

4

Promotional Sunset Plan

How should the brand restructure its promotional strategy to protect margins?

```sql
SELECT
    Master_Customer_Segment,
    COUNT(*) AS customer_count,
    ROUND(AVG(Promo_Dependency_Score), 2) AS avg_promo_dependency,
    ROUND(AVG(Purchase_Amount_USD), 2) AS avg_spend,
    ROUND(AVG(Loyalty_Score_A), 2) AS behavioral_loyalty,
    CASE
        WHEN AVG(Promo_Dependency_Score) >= 0.75 AND AVG(Loyalty_Score_A) < 0.50
            THEN 'Sunset Discounts – Low loyalty risk'
        WHEN AVG(Promo_Dependency_Score) >= 0.50 AND AVG(Loyalty_Score_A) >= 0.50
            THEN 'Reduce Gradually – Monitor retention'
        WHEN AVG(Promo_Dependency_Score) < 0.40
            THEN 'Protect – Organically loyal, no discounts needed'
        ELSE 'Review Case by Case'
    END AS promo_recommendation
FROM d2c_customers
GROUP BY Master_Customer_Segment
ORDER BY avg_promo_dependency DESC;
```

Query Result:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Master_Customer_Segment	customer_count	avg_promo_dependency	avg_spend	behavioral_loyalty	promo_recommendation
Discount Survivors	1640	1.00	59.12	0.5	Sunset Discounts — Low loyalty risk
Premium Loyalists	1420	0.00	60.19	0.8	Protect — Organically loyal, no discounts needed
Moderate Customers	472	0.00	61.29	0.6	Protect — Organically loyal, no discounts needed
Emotionally Loyal	331	0.00	58.21	0.67	Protect — Organically loyal, no discounts needed

Key Insights:

- Segments with:
  - High promo dependency
  - Low loyalty

  were classified as safe for discount reduction.
- Loyal customers with low promo dependency should be protected from unnecessary discounts.
- CASE logic helped automate customer treatment recommendations.

Business Understanding:

Three strategic groups emerged:

- Sunset Discounts → low loyalty + high promo dependency
- Reduce Gradually → medium loyalty + promo dependency
- Protect Loyal Customers → naturally loyal buyers

This helps improve profit margins without damaging retention.

5

Ideal Customer Profile

What does the brand's best customer actually look like?

```sql
SELECT
    Gender,
    ROUND(AVG(Age), 1) AS avg_age,
    Location,
    Category,
    Subscription_Status,
    Season,
    ROUND(AVG(Purchase_Amount_USD), 2) AS avg_spend,
    ROUND(AVG(Previous_Purchases), 1) AS avg_purchases,
    ROUND(AVG(Loyalty_Score_A), 2) AS behavioral_loyalty,
    COUNT(*) AS customer_count
FROM d2c_customers
WHERE Master_Customer_Segment = 'Premium Loyalists'
GROUP BY Gender, Location, Category, Subscription_Status, Season
ORDER BY avg_spend DESC, behavioral_loyalty DESC
LIMIT 20;
```

Query Result:

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

Gender	avg_age	Location	Category	Subscription_Status	Season	avg_spend	avg_purchases	behavioral_loyalty	customer_count
Male	46.0	Arizona	Footwear	No	Summer	100.00	49.0	0.98	1
Female	22.0	Hawaii	Outerwear	No	Fall	100.00	42.0	0.87	1
Female	28.0	Arizona	Outerwear	No	Fall	100.00	48.0	0.85	1
Female	45.0	New Mexico	Clothing	No	Spring	100.00	33.0	0.85	1
Male	20.0	North Dakota	Outerwear	No	Summer	100.00	42.0	0.82	1

Key Insights:

- Premium loyal customers commonly showed:
  - High purchase frequency
  - High loyalty scores
  - High average spending
- Specific combinations of:

  Age, Gender, Category, Location, Season

  helped identify the strongest customer profile.

Business Understanding:

The company can now build:

- Highly targeted campaigns
- Better ad audiences
- Personalized retention strategies

This query converts raw customer data into a clear marketing persona.

Season x Category - Tenure Analysis

(Additional Analysis - Beyond Scope)

Which seasons and categories attract new vs. established repeat customers?

```
SELECT
    Season,
    Category,
    ROUND(AVG(Previous_Purchases), 1) AS avg_previous_purchases,
    ROUND(AVG(Promo_Dependency_Score), 2) AS avg_promo_dependency,
    COUNT(*) AS customer_count,
    CASE
        WHEN AVG(Previous_Purchases) < 10 THEN 'Lower Tenure'
        ELSE 'Higher Tenure'
    END AS tenure_group
FROM d2c_customers
GROUP BY Season, Category
ORDER BY avg_previous_purchases DESC;
```

Query Result:

Season	Category	avg_previous_purchases	avg_promo_dependency	customer_count	tenure_group
Winter	Footwear	26.8	0.40	139	Higher Tenure
Summer	Accessories	26.7	0.42	309	Higher Tenure
Spring	Outerwear	26.1	0.40	80	Higher Tenure
Winter	Clothing	26.0	0.42	443	Higher Tenure
Spring	Accessories	25.7	0.46	298	Higher Tenure

Key Insights:

- Categories with higher previous purchases indicate stronger long-term retention.
- Some seasonal-category combinations showed:
    - Higher repeat purchase behavior
    - Lower promo dependency
- Customers were segmented into:
    - Lower Tenure
    - Higher Tenure

Business Understanding:

The company can:

- Identify products that create repeat buyers.
- Focus retention campaigns around strong seasonal categories.
- Predict which product categories build long-term customer relationships.

Executive Summary

- Performed SQL-based customer segmentation and behavioral analysis on a D2C fashion retail dataset to identify loyalty patterns, promotional dependency, and customer retention opportunities.
- Used aggregation queries, CASE statements, conditional logic, and grouping techniques to analyze customer behavior across segments, locations, seasons, and product categories.
- Identified that **Premium Loyalists** generate organic purchases with minimal discount dependency, while **Discount Survivors** rely heavily on promotions for conversions.
- Discovered that customer value tiers alone are not strong indicators of loyalty, highlighting the importance of behavioral segmentation over revenue-based segmentation.
- Analyzed geographic demand patterns to identify regions with strong organic customer engagement and lower promotional reliance.
- Developed a SQL-based promotional recommendation framework to classify customer segments into:
    - Protect Loyal Customers
    - Reduce Discounts Gradually
    - Sunset Discount Dependency

- Built an Ideal Customer Profile (ICP) by analyzing demographic, seasonal, geographic, and purchasing behavior of high-loyalty customers.
- Generated actionable insights to support customer retention strategies, promotional optimization, and data-driven marketing decisions using SQL analytics
- The project intentionally prioritized customer intelligence, retention realism, and founder-oriented decision-making over unnecessary machine learning complexity.

Recommended Next Steps

Priority	Action	Segment / Market	Expected Impact
HIGH	Sunset discounts for highly promo-dependent customers	Discount Survivors	Improve profit margins and reduce discount dependency
HIGH	Launch VIP rewards and early-access programs	Premium Loyalists	Increase retention, referrals, and customer lifetime value
HIGH	Increase marketing investment in high organic-demand regions	Arizona, Alaska	Strengthen brand-led growth and improve acquisition efficiency
MEDIUM	Rebuild customer targeting using behavioral segmentation	All Customer Tiers	Improve campaign personalization and targeting accuracy
MEDIUM	Optimize seasonal product promotions strategically	All Seasons & Categories	Reduce unnecessary promotions and improve revenue quality
LOW	Monitor emotionally loyal niche customers separately	Emotionally Loyal Segment	Preserve long-term loyalty and niche customer retention